Electrical and Electronics Engineering and Robotics

ELEC345 High Speed Communications

The Wireless Environment

Dr Toby Whitley

UNIVERSITY OF PLYMOUTH

Today's objectives, to understand…

- Multipath?
- Fading, fast and slow.
- Rayleigh and Rician fading.
- Probability models for slow and fast fading.
- Time dispersion and the idea of a time varying impulse response
- Channel properties caused by fading

MULTIPATH

Signal lost

Spoiling mobile communications for years.

Multipath

- Refers to the propagation of energy from transmitter to receiver via multiple paths
- Direct path is the straight line path from transmitter to receiver and in the absence of obstructions is synonymous with the line-of-sight (LOS) path
- Other paths arise due to

  Reflection

  Diffraction

  Scattering

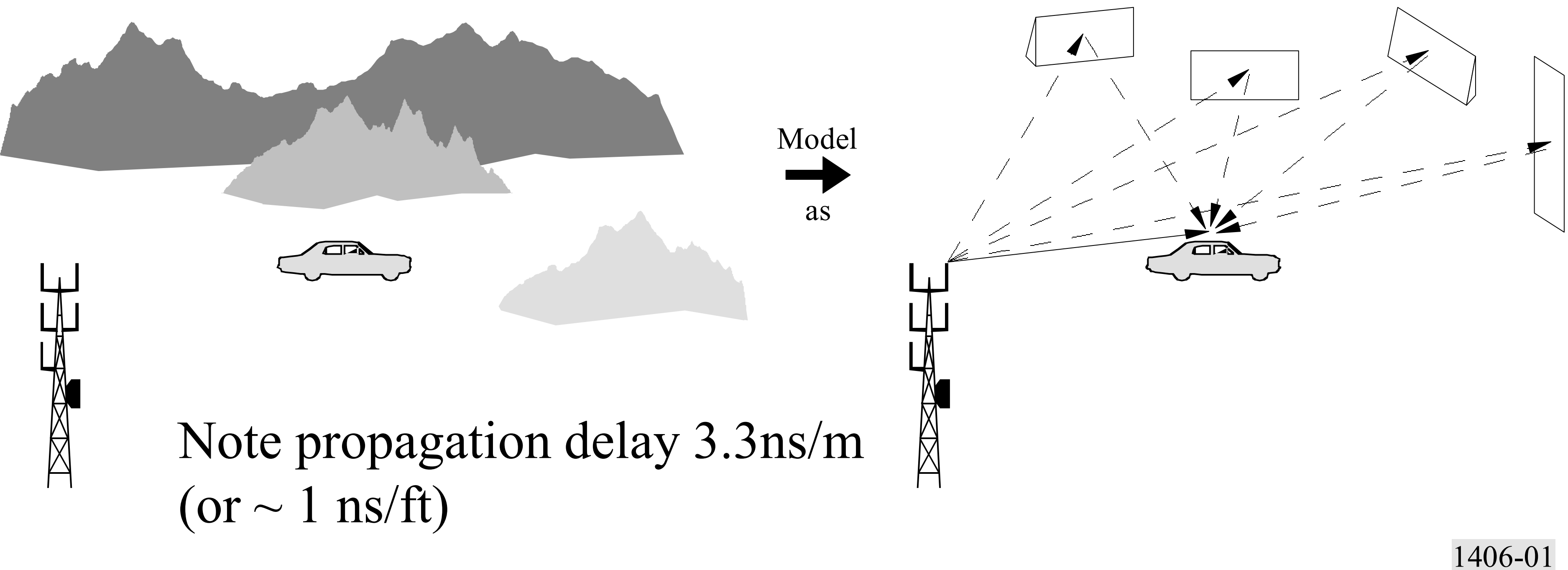
WITH PLYMOUTH UNIVERSITY

Multipath

- ***Multipath propagation results in a received signal that is dispersed in time, for digital signalling this leads to a form of ISI whereby data samples are corrupted by the responses of neighbouring data symbols.***
- The term scattering is used
    - (loosely) to denote reflection and/or diffraction
    - (commonly) to denote reflection from electrically small objects (size less than several wavelengths)
    - (strictly) to define the scattered field
        - equal to the actual field (i.e. the field in the presence of a scattering object) minus the incident field (i.e. field in the absence of the scattering object

Rural Multipath propagation

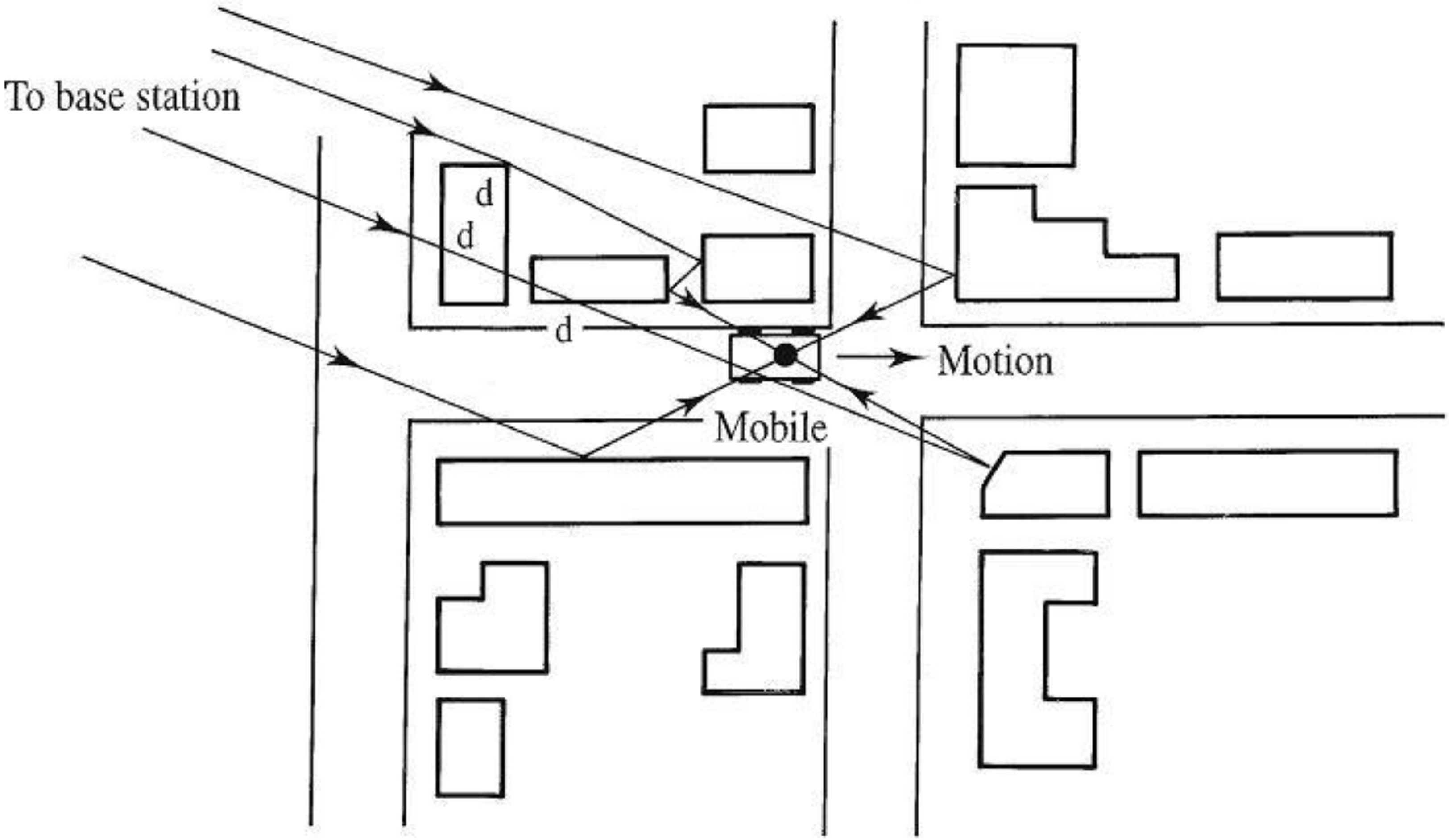
Long excess delays possible, few (μs) for mountainous rural areas, and for flat rural areas.



Note propagation delay 3.3ns/m (or ~ 1 ns/ft)

WITH PLYMOUTH UNIVERSITY

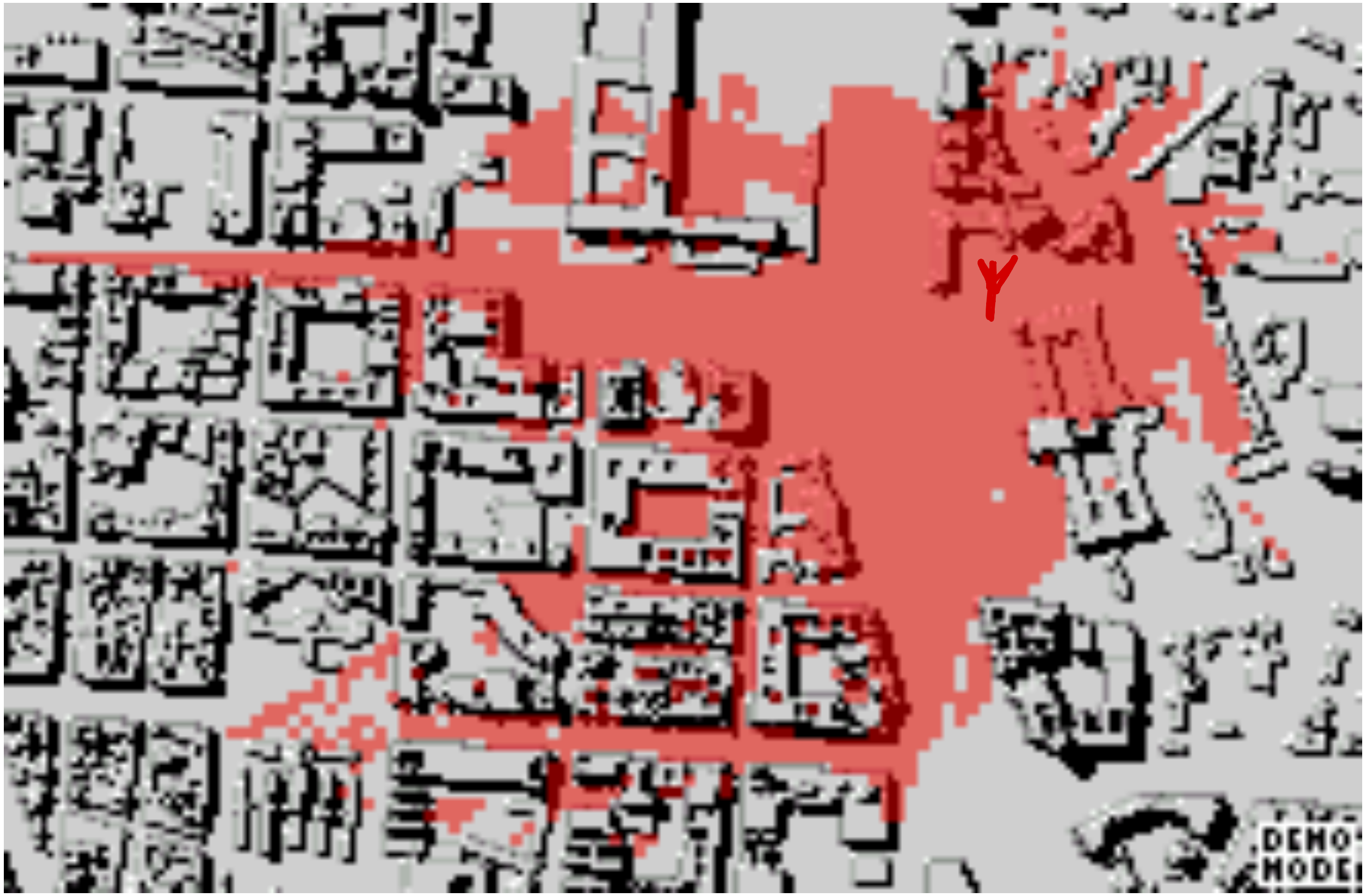
Urban Multipath propagation



- Multipath origin of Doppler shift, fading and dispersion in a mobile radio channel.

WITH PLYMOUTH UNIVERSITY

A real urban Multipath propagation scenario



Disadvantages of the mobile channel

- Disadvantages of a mobile radio channel are:
  - Doppler shifts in the carrier frequency
  - Slow spatial fading
  - Rapid spatial fading
  - Temporal fading
  - Frequency selective fading
  - Time dispersion (due to Multipath, i.e. ISI)
  - Time variation of channel characteristics (due to movement, i.e. mobility)

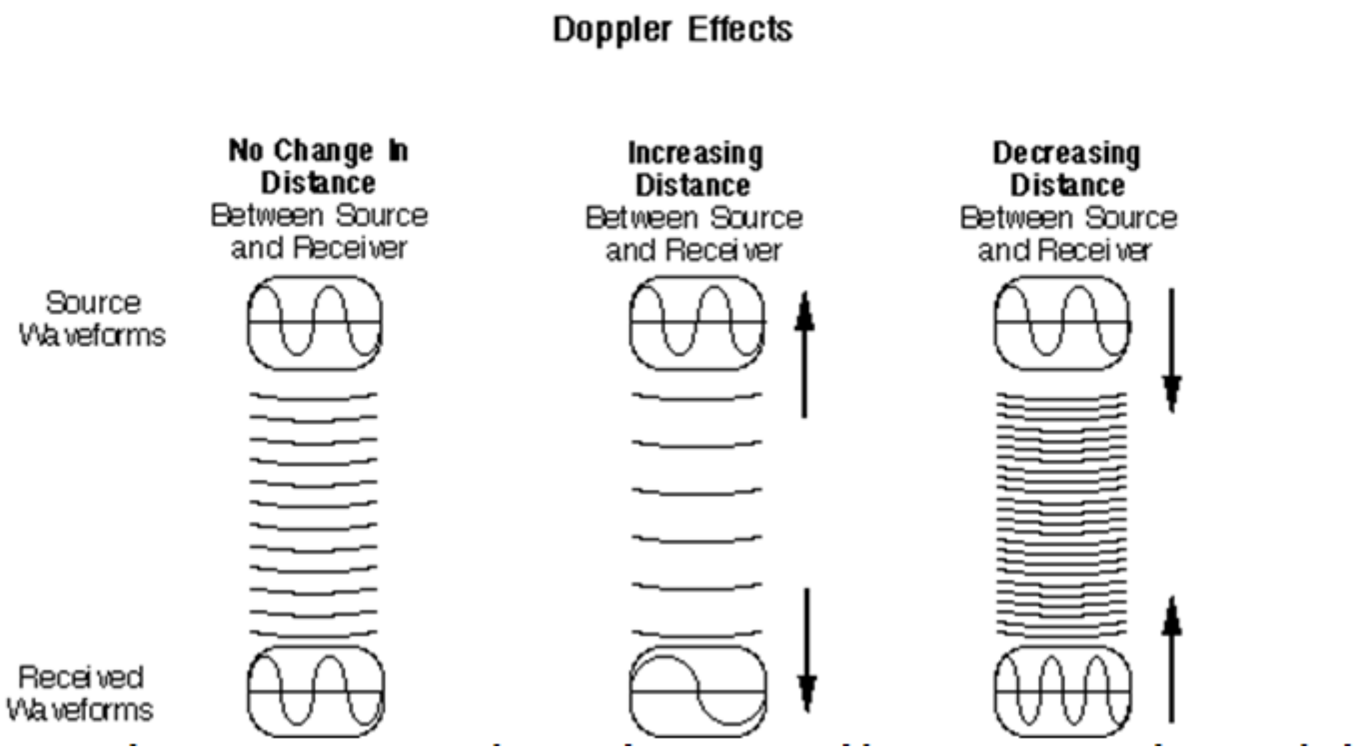
Propagation effects in the mobile channel

- Doppler shifts in carrier (Due to relative motion between terminals)
  - Doppler shift happens when a user (or reflectors in its environment) is moving, the user's velocity causes a shift in the frequency of the signal transmitted along each signal path. Signals travelling along different paths can have different Doppler shifts, corresponding to different rates of change in phase.
- Doppler shifts in carrier $\left(f_d = \frac{v}{\lambda} = \frac{v}{c} f_{Hz} \right)$

Where f_d is Doppler frequency in Hz, v is the relative velocity in the direction of propagation, f_{Hz} is the transmitted frequency in Hz and c is the speed of light (3×10^8 m/s). **Example:** At 900MHz about 1.2km/hr velocity produces 1Hz Doppler shift.

WITH PLYMOUTH UNIVERSITY

Doppler Effects



WITH PLYMOUTH UNIVERSITY

Propagation effects in the mobile channel

- **The Coherence Time (*T_c*) is a measure of the minimum time required for the magnitude change of the channel to become de-correlated from its previous value**

- The **coherence time** of the channel is inversely related to a quantity known as the **Doppler spread** of the channel.

WITH PLYMOUTH UNIVERSITY

Propagation effects in the mobile channel

- Channels with a large Doppler spread have signal components that are each changing independently in phase over time. Since fading depends on whether signal components add constructively or destructively, such channels have a very short coherence time. In general, coherence time is inversely related to Doppler spread, typically expressed as:

$$T_c = \frac{k}{D_s} \qquad Tc \sim 0.4/fd$$

where *Tc* is the coherence time, *Ds* is the Doppler spread, and *k* is a constant taking on values in the range of 0.25 to 0.5.

WITH PLYMOUTH UNIVERSITY

Propagation effects in the mobile channel

- Slow spatial fading (urban areas)

  arises when the coherence time of the channel is large relative to the delay constraint of the channel. In this case, the amplitude and phase change imposed by the channel can be considered roughly constant over the period of use

  Due principally to topographical shadowing/diffraction

  Also called large-scale fading

Propagation effects in the mobile channel

- Fast spatial fading (multipath propagation)

Due to constructive and destructive interference between signals arising from multiple paths

Also called small-scale or multipath fading

Fast fading occurs when the coherence time of the channel is small relative to the delay constraint of the channel. In this case, the amplitude and phase change imposed by the channel varies considerably over the period of use.

In a fast-fading channel, the transmitter may take advantage of the variations in the channel conditions using diversity to help increase robustness of the communication to a temporary deep fade.

Effects of multipath propagation on the mobile channel

- Temporal fading
  - Due, principally, to mobile's motion through spatially varying field but static terminal may suffer temporal fading due to movement of scatterers
- Time dispersion
  - Due to range of delays introduced by multiple paths
- Frequency selective fading

  This is a direct consequence of time dispersion arising due to multipath propagation
  - Some frequencies may be faded more than others if signals are broadband
- Time variation of channel characteristics
  - Due, principally, to movement of the mobile terminal

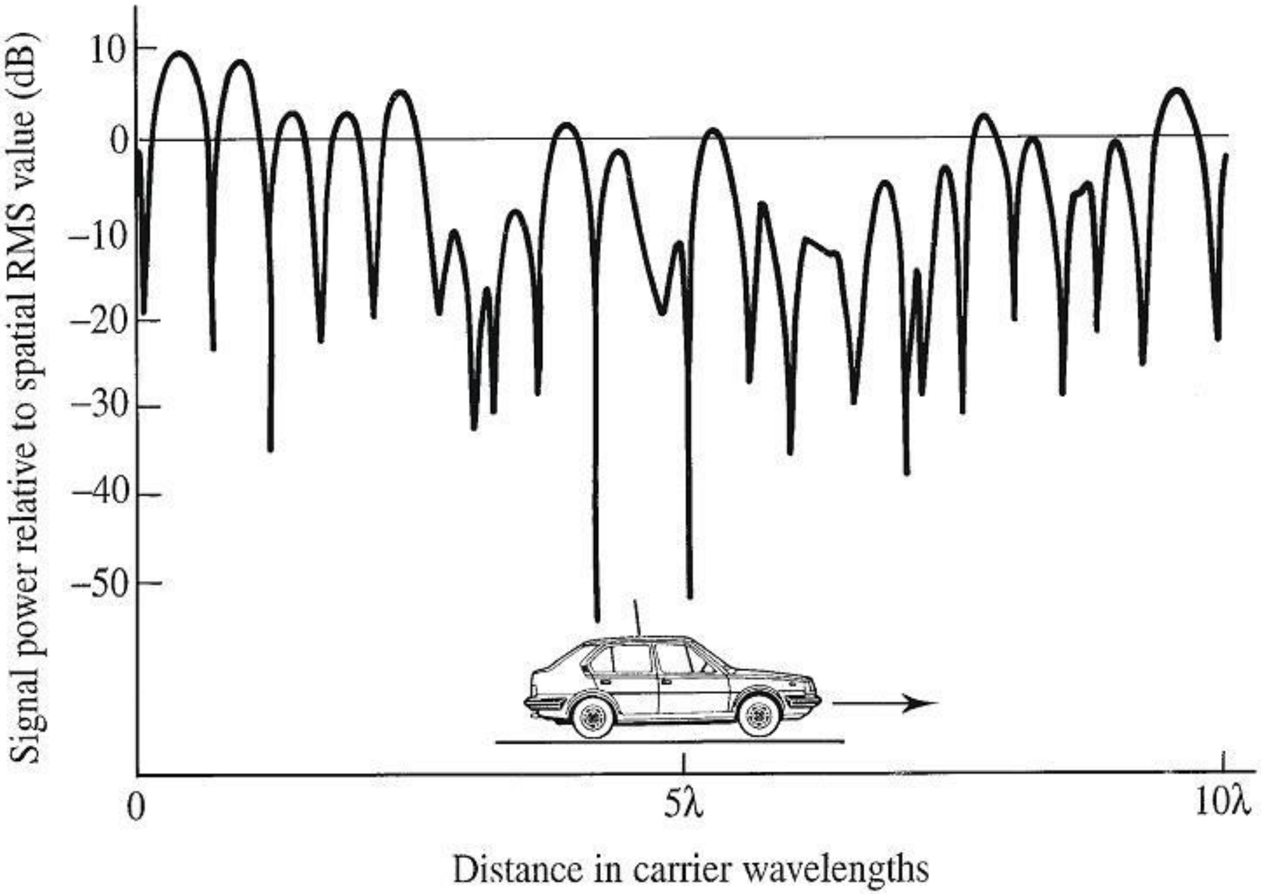
Relationship between multipath effects

- Not all the above effects are independent
- e.g. frequency selective fading is a frequency domain manifestation of time dispersion
- Also notice the relationship between fading effect and the frequency content of the signal

Typical multipath fading power profile

-Instant. received power for a mobile w.r.t. local spatial average or mean power level. This is spatial fading due to mobile motion.

-If the reflectors are moving objects, i.e. vehicles, then temporal fading will result.

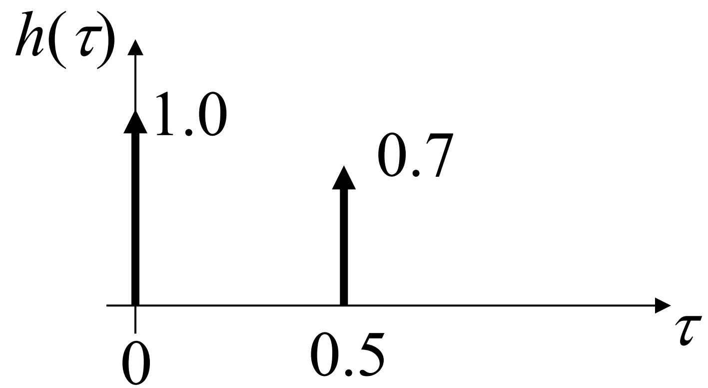


WITH PLYMOUTH UNIVERSITY

Multipath fading

- Caused by constructive and destructive interference between multipath signals
- Takes place on (spatial) scale of wavelength
  - Note approx half wavelength fading period in last slide
- Converted to temporal fading by movement of receiver through spatial field
- True temporal fading (at a fixed point) caused by
  - Movement of transmitter
  - Movement of scatterers (i.e. changing environment)
- Also called small-scale or fast fading

Two path example



Assume a two ray multipath channel :

$$h(\tau) = \delta(\tau) + 0.7\delta(\tau - 0.5)$$

The frequency response is then given by FT $\{h(\tau)\}$, i.e:

$$H(f) = 1 + 0.7e^{-j2\pi f\tau}$$

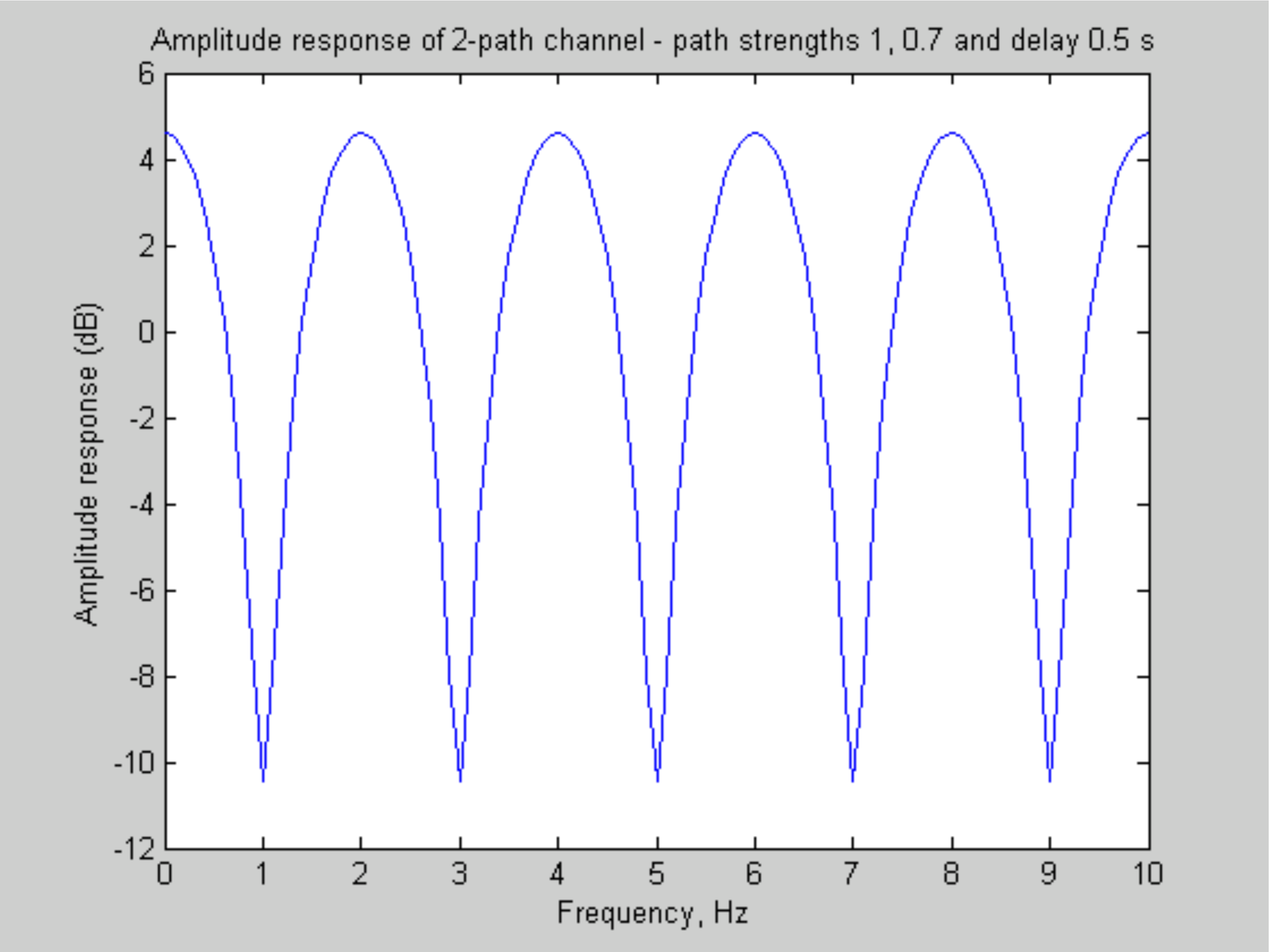
and the amplitude response is given by :

$$|H(f)| = \left|1 + 0.7e^{-j2\pi f\tau}\right|$$

[Note delay (0.5 s) is illustrative only]

WITH PLYMOUTH UNIVERSITY

Two path example



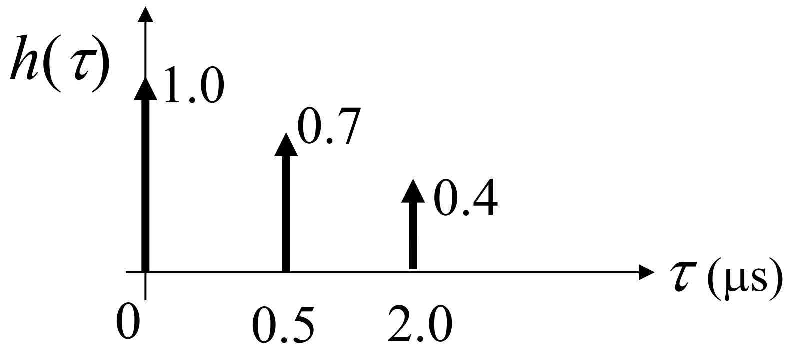
Three path example

- Three ray channel with impulse response:

$$h(\tau) = 1 + 0.7\delta(\tau - 0.5\times10^{-6}) + 0.4\delta(\tau - 2.0\times10^{-6})$$

- Has the frequency response:

$$H(f) = 1 + 0.7e^{-j2\pi 0.5\times10^{-6} f} + 0.4e^{-j2\pi 2.0\times10^{-6} f}$$



[Delays of 0.5 and 2 μs are plausible for macrocell]

WITH PLYMOUTH UNIVERSITY

Three path example

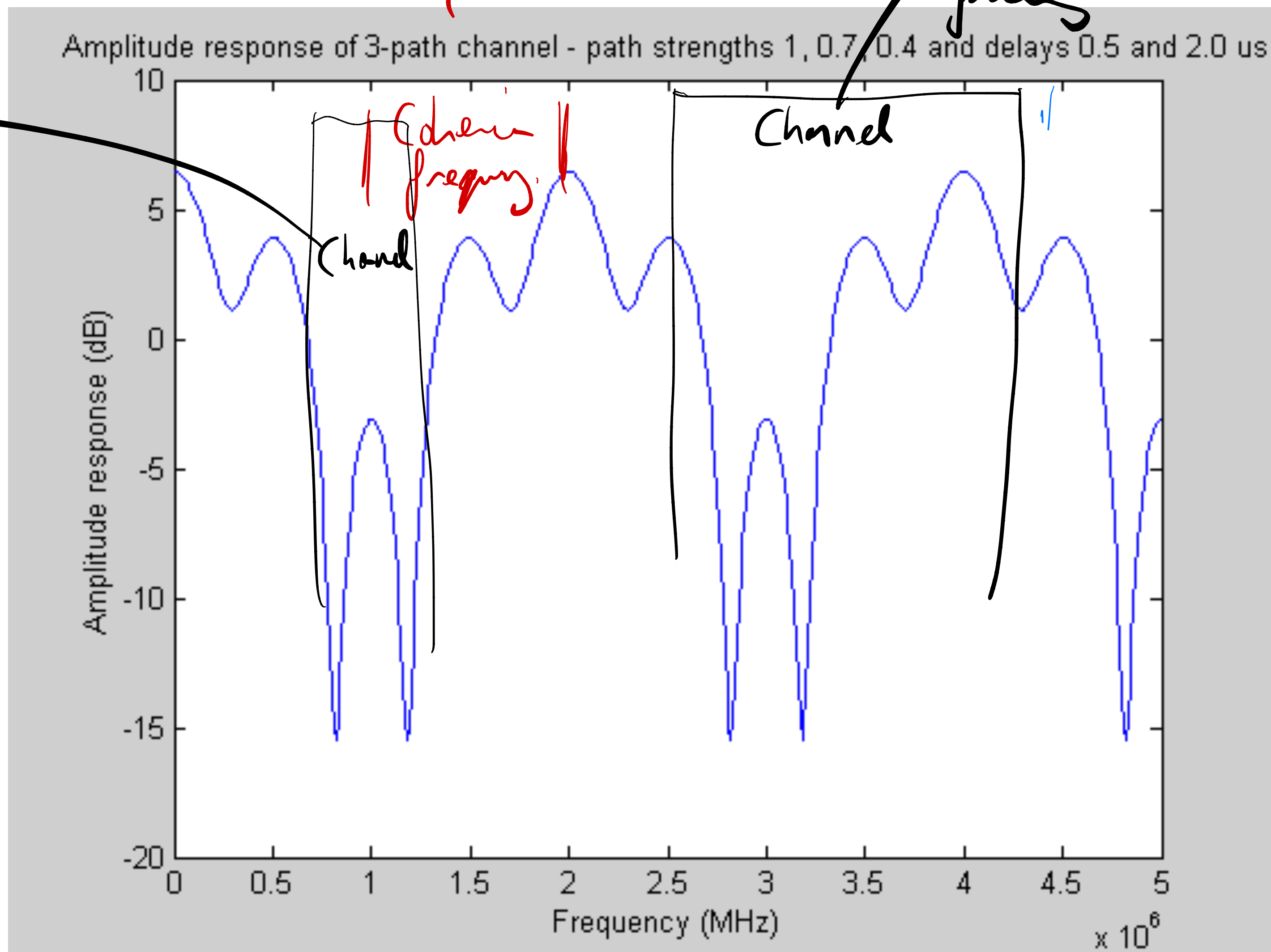
Broadband ≥ cf.

Frequency selective fading

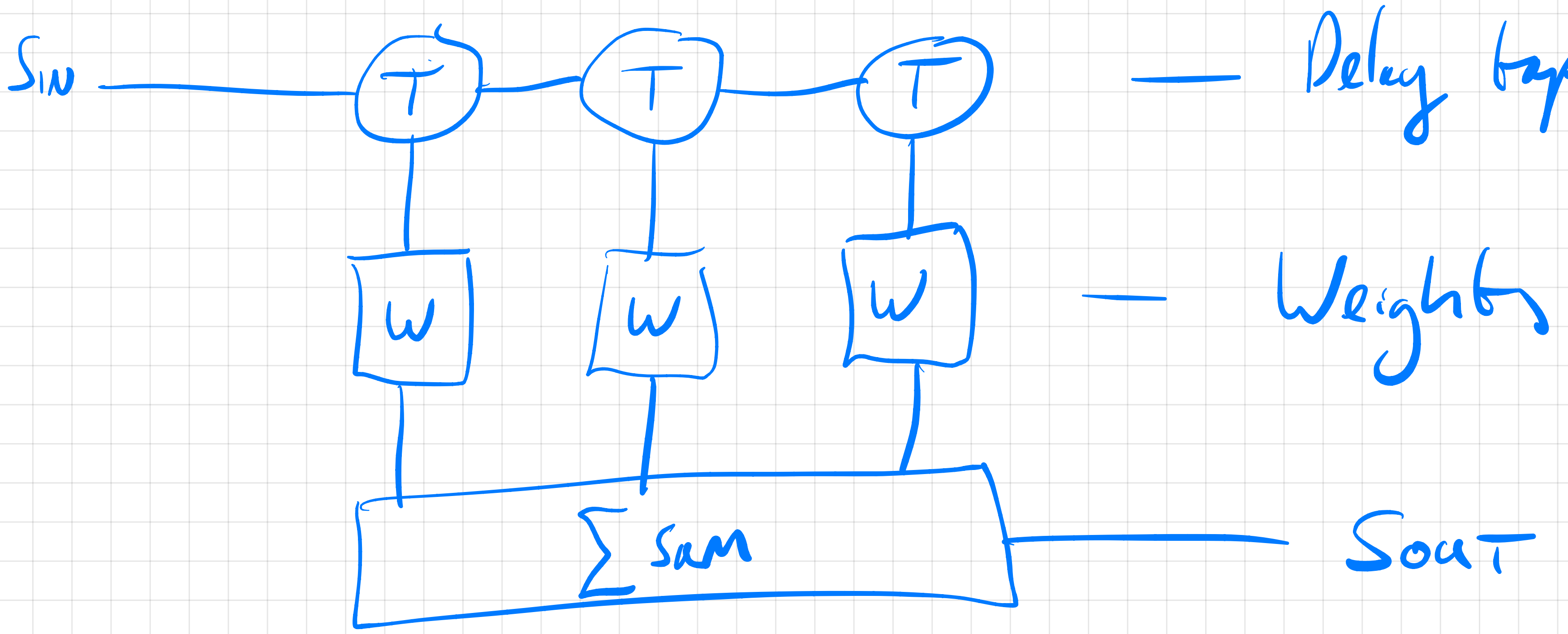
Decorrelated

channel < cf

narrowband channel



Digital Filter — Can simulate a channel.



Filter does frequency selective fading in hardware.

Frequency selective fading

- Notice that the amplitude responses in the previous examples strongly attenuates certain (selected) frequencies
- This is called frequency selective fading
- Systems that have narrow bandwidth with respect to the frequency selectivity are said to be frequency non-selective channels and suffer from flat fading
- Frequency selectivity is a direct consequence of time dispersion arising due to multipath propagation
- Frequency-selective fading channels are also dispersive, in that the signal energy associated with each symbol is spread out in time. This causes transmitted symbols that are adjacent in time to interfere with each other.

WITH PLYMOUTH UNIVERSITY

Flat fading vs frequency selective fading

- The **coherence bandwidth** measures the minimum separation in frequency after which two signals will experience uncorrelated fading, and roughly equals the reciprocal of the RMS delay spread
- In **flat fading**, the coherence bandwidth of the channel is larger than the bandwidth of the signal. Therefore, all frequency components of the signal will experience the same magnitude of fading.
- In **frequency-selective fading**, the coherence bandwidth of the channel is smaller than the bandwidth of the signal. Different frequency components of the signal therefore experience de-correlated fading.

Summary

- Fast (or multipath) fading occurs on a spatial scale of wavelengths and can usually be modelled either as

  a bivariate Gaussian random process (with uniform phase distribution and Rayleigh amplitude distribution)

  A bivariate Gaussian process plus a constant (resulting in a Ricean amplitude distribution and a phase distribution peaked around the phase of the constant

- Slow fading (or shadowing) results from the cascading (multiplying together) of several random shadowing processes

Example questions

What is coherence time.

What does de correlated mean

In terms of coherence bandwidth, what makes a narrow band or broadband channel.

What is a flat fade

Over what distance would you expect to move to see changes in recieved signal due to multipath.

What is fast fading?